

GEM Metadata Creation Guidelines

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Hakai Catalogue

In order to make your data discoverable and accessible to the public, and ensure you can be properly credited for the work you have done, you will need to create a metadata record. Metadata essentially being 'data about data', this metadata record will provide additional context for your dataset, and it will be stored in the [Hakai Catalogue](#). Each participant will have to create their own metadata record so that proper attribution can be given. A metadata record does not *store* the data, rather, it points users to the GitHub repository where the data is located.

Metadata Entry Tool

You can use the [Hakai Metadata Entry Tool](#) to fill out your metadata. It ensures your metadata follows the international ISO-19115 standard, allowing it to be found by other catalogues and search engines, such as Google Search. It will help make your work even more discoverable. To use the tool, you will have to sign in - you can only do so if you have a Google account (e.g., gmail). If you don't have a Google account, please let me know by sending me an email at either data@hakai.org or tim.vanderstap@hakai.org and I can help you further.

Some context: This tool was used by the Canadian Integrated Ocean Observing System to populate metadata records for a catalogue that needed to be useful for ocean practitioners across Canada. Therefore, it's bilingual. You can see this at various stages, where you can translate your metadata from English to French with a translate button, or where you can select the primary language of the record to be French rather than English. **Translate a field to French whenever you have that option.**

At Hakai, we have made slight changes to the form to accommodate the data we collect, and the research programs that we have. Therefore, there will be a few fields throughout the Metadata Entry Tool that will not be applicable to the GEM Initiative.

Once signed in, go to 'My records' in the top left corner, and select "New record".

Tool Layout

At the top of the screen there are 8 tabs, ranging from Start to Submit. Below the tabs, you will see a progress bar. When this bar is at 100%, which happens by filling out the **required** fields, you will be able to submit your metadata record for review. The required fields are marked with a red asterisk. Additional information can be found at the top of this page. Once you have added a title, you can save your progress by selecting the save icon in the bottom right corner and come back to it at a later time.

Start

Do not populate the "Import existing metadata from DOI" field. This is used to import existing metadata from a DOI and currently does not apply to you.

You need to provide a title for your dataset and translate it to French using the 'Translate' button. I would recommend including 'Global Eutrophication Monitoring' in your title, as this will make it easier to search for this metadata record in the catalogue once it's published.

The resource type can stay 'Dataset'.

The theme of the record should be set to 'Oceanographic'.

Keep the primary language of the dataset as English.

In the next field generate a *draft DOI*. You will first have to save your progress. A DOI is a Digital Object Identifier, a unique string that will ensure long-term preservation and accessibility of your dataset. The DOI is minted through DataCite, with the Hakai Institute being a member of the Canadian DataCite consortium. **After filling out the rest of the metadata, you will have to click 'update DOI' before submitting.**

Finally, in this tab you can share editing access with another user, but you will only be able to find other users that have signed in previously, or created / contributed to a metadata record.

Before moving on to the next section, make sure to save your progress by clicking the save icon!

Resource Identification

Under the 'Resource Identification' tab, you can skip the first field, as your work is not part of a Hakai project or program.

Next, you can fill out an abstract for your dataset. Try to keep your abstract short and to the point, touching on the **what**, **when**, **where**, and **why** of your data.

Essential Ocean Variable (EOV) field: select the EOVs that apply to your dataset. For context, the Global Ocean Observing System ([GOOS](#)) - has defined 'Essential Ocean Variables' (EOVs) as primary variables which can be feasibly measured and be impactful, providing a good assessment of the state of our oceans.

In addition, you can also choose more specific keywords that fit your dataset. When adding keywords, make sure to press 'Add' following each keyword or phrase. Aim for 5 - 10 keywords.

Next, select the status of the dataset. The metadata record should be created and published prior to or during the active phase of data collection and the status should be set to "on-going". When your project is complete and you are no longer collecting data for the GEM program, the status will need to be changed to "complete".

In the next three fields you can provide information on:

- Start date (first day data was collected);
- End date (last day data was collected - left blank as on-going);
- Publication date - this will be the date the GitHub repository is made public, 2025-12-31.

The version of your dataset will initially be 1.0 upon publication. If you need to update your metadata record to reflect future changes in data collection (e.g. new sampling areas, stations, methods or instruments etc.), then a new version can be created. The date a new version is published will be the revision date.

For licensing, select Creative Commons Attribution 4.0 Attribution, which is [CC-BY-4](#). Under these terms and conditions, downstream users of your data can freely use, modify and share your data, while having to provide appropriate attribution to the creators of the dataset.

Finally, in this tab you can add any limitations affecting your dataset.

Before moving on to the next section, be sure to save your progress again!

Taxonomic Classification

Indicate that this dataset does not contain any taxonomic classification.

Spatial

Here you provide the spatial extent of your study area as either a bounding box (i.e. the minimum and maximum latitude and longitude coordinates in decimal degrees), a drawn polygon on the map or the coordinates of a polygon that covers the study area. If you are providing polygon coordinates, they must start and end with the same point. (e.g, 48,-128 56,-133 56,-147 48,-128).

Also, provide a description of the geographic extent - this is free text.

Finally, enter the vertical extent that your data covers. Select “Depth Positive” indicating that your depth values are below the surface of the water (e.g. 5 indicates 5 meters *below* the surface water level).

Contact

This is where you indicate how you want users to cite your work. To add a new contact, select “Add new item”. Add the roles that this contact has - there will have to be **at least** one Metadata Custodian and one Data Owner. These can be the same person. You can select more roles by clicking ‘Show/Hide more role options’. Hover over the roles to see what their definitions are.

For each contact, you need to provide information on their affiliation. Some affiliations are registered on the [Research Organization Registry](#), or ROR. To see if an affiliation is registered, type the name of the institution or organization in the top field. Selecting ROR registered affiliation will populate the majority of the remaining organization fields. If a ROR does not exist, add as much information about the organization as you can, such as the name, address, country and email.

Below, add information about the individual. If the individual has an [ORCID](#), providing their identifier will automatically populate their first and last name. Otherwise, fill in the fields manually. Please do this for all the contacts that you think should be included in the recommended citation. Who should be included in the citation is up to the lead author or Principle Investigator (PI).

The order of the authors listed in the citation is determined by the order in which they’re listed in this record. You can change this order by dragging contacts to their respective positions using the = sign next to their name. The recommended citation will appear in the metadata record and is the way in which most scientific works get credited, so it’s important that you are happy with the order (again, this is generally decided by the lead author or PI).

Finally, add the ‘Hakai Institute’ with the role of ‘Publisher’, given that this metadata record will be included in the Hakai Catalogue. To add the Hakai Institute, create a new contact and search for the Hakai Institute in the ROR field. You’ll notice that the Publisher doesn’t get added in the list of authors, but at the end of the citation (in the Publisher field).

Data and Documentation

Under Resource, select “Add Resource” and then add the link to the GitHub repository where your data is stored and provide a title and brief description of what the repository contains.

As you scroll down, you can see there are separate sections for “Related Works” and “Lineage”. In the future, the “Related Works” section will be the most relevant as you will need to update this section (Lineage is not important for GEM). This is where you will add resources, publications, reports, or other metadata records that are derived from, or related to, your data as

they become available. When you click “Add Resource”, you need to provide the title of the related work, the identifier (this will likely be either a URL or a DOI), as well as what the relationship is between the primary resource (your GEM dataset) and this related work. For example, if your data was published in a journal, you could add the relationship “Is Published In”, with a link to the publication.

Platform

Include information on the platform(s) and the instruments used. The platform is anything used in data collection that has the instrument(s) attached to it.

Select “lowered unmanned submersible”. Add your GEM Kit number as the unique platform ID or code.

Scrolling down, add information for each instrument. The name of the instrument should be included under Instrument ID:

- PME miniDOT,
- CTD-diver

For the Aquafluor and DR1900, add to the description of these instruments that they were used in the lab to analyze collected water samples.

Submit

Once you have filled out all the information, scroll back to the [Start](#) tab and click ‘**update DOI**’. The final tab ‘Submit’ is where you submit your metadata for review. You can only submit your metadata record if the progress bar is at a 100%, which will only happen if you have filled out all the *required* fields. If you have not done that, you will see an error message on this page indicating which fields you have not completed yet.

When you submit a metadata record, a GitHub ticket will be created [here](#), and I will do a quick review to make sure that your metadata is correct (e.g., is there useful information in the abstract, is an email provided for points of contact, etc). Once this is all confirmed, we can publish the record, and it will be discoverable on the Hakai Catalogue along with a unique DOI that users can reference, making sure that you get the credit you deserve for the work that you have put in!

I hope this helps – if you have any questions, please feel free to send me an email at tim.vanderstap@hakai.org!